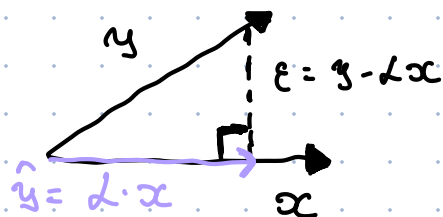


QR - разложение, ортогонализация и проекции

① Проекция на вектор

$$y = \alpha \cdot x + (y - \alpha x) = \alpha x + \varepsilon$$



$$\varepsilon \perp x$$

$$y - \alpha x \perp x$$

$$\langle y - \alpha x, x \rangle = 0$$

$$\langle y, x \rangle - \alpha \langle x, x \rangle = 0$$

$$\alpha = \frac{\langle y, x \rangle}{\|x\|^2}$$

$$\text{Proj}_x y = \alpha \cdot x$$

$$\text{Proj}_x y = \arg \min_z \|y - z\|_2$$

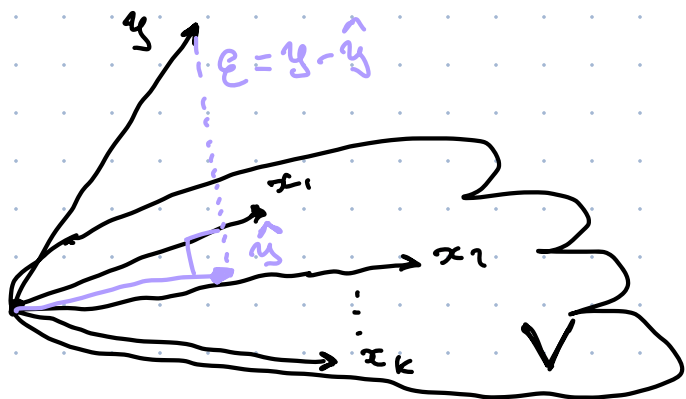
$$\sqrt{\sum_{i=1}^n (y_i - z_i)^2} \rightarrow \min_{z_1, \dots, z_n}$$

$$\sqrt{\sum_{i=1}^n (y_i - \alpha x_i)^2} \rightarrow \min_{\alpha}$$

Получим обычную
линейную регрессию

② Расширяем!

y $x_1, x_2, \dots, x_k$



$$V = \text{span} \{x_1, \dots, x_k\}$$

$$\hat{y} = \underbrace{\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_k x_k}_{\alpha \beta}$$

$$\hat{y} \in \text{span} \{x_1, \dots, x_k\}$$

$$\hat{y} = \text{Proj}_V y$$

$$\varepsilon = y - \hat{y} \perp V$$

Лин. рез. - тоже проекция

$$\varepsilon \perp x_j$$

$$\langle \varepsilon, x_j \rangle = 0$$

$$X^T \varepsilon = \begin{pmatrix} \langle \varepsilon, x_1 \rangle \\ \langle \varepsilon, x_2 \rangle \\ \vdots \\ \langle \varepsilon, x_k \rangle \end{pmatrix} = 0$$

$$X^T (y - X\beta) = 0$$

$$X^T y = X^T X \beta$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$\hat{y} = X \hat{\beta} = X (X^T X)^{-1} X^T y$$

H hat matrix
матрица-шляпница

③ Проекция в ортонорм. базисе

Опр. ортонормированный базис

$q_1, \dots, q_k$ - ортонорм. базиса

$$1) \forall i \neq j \quad q_i \perp q_j$$

$$2) \|q_i\| = 1$$

$$\hat{y} = \text{proj}_V y = \sum_{j=1}^k \langle y, q_j \rangle \cdot q_j$$

$$y = \beta_1 q_1 + \dots + \beta_k q_k$$

$$\langle y, q_1 \rangle = \beta_1 \underbrace{\langle q_1, q_1 \rangle}_{=1} + \beta_2 \underbrace{\langle q_2, q_1 \rangle}_{=0} + \dots + \beta_k \underbrace{\langle q_k, q_1 \rangle}_{=0}$$

④ Ортогонализация Грамма - Шмидта

$$v_1, v_2, v_3$$

$$u_1 = v_1$$



$$u_2 = v_2 - \text{proj}_{u_1} v_2$$

$$u_2 \perp u_1$$

$$u_3 = v_3 - \text{proj}_{u_1} v_3 - \text{proj}_{u_2} v_3$$

$$u_3 \perp u_2 \quad u_3 \perp u_1$$

q_1, q_2, q_3 - ортонорм. баз.

$$q_1 = \frac{u_1}{\|u_1\|}$$

$$q_2 = \frac{u_2}{\|u_2\|}$$

$$q_3 = \frac{u_3}{\|u_3\|}$$

$$\begin{aligned} \text{span}\{v_1, v_2, v_3\} &= \\ &= \text{span}\{q_1, q_2, q_3\} \end{aligned}$$

Упражнение

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$u_1 = v_1$$

$$v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$u_2 = v_2 - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} \cdot u_1 =$$

$$v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{2}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2/3 \\ 1/3 \\ 1/3 \end{pmatrix}$$

$$\langle v_2, v_1 \rangle = 2$$

$$\langle u_1, u_1 \rangle = \langle v_1, v_1 \rangle = \|u_1\|^2 = 3$$

$$\begin{aligned} \langle u_2, u_1 \rangle &= 0 \\ \|u_2\| &= \sqrt{6}/3 \end{aligned}$$

$$u_3 = v_3 - \frac{\langle v_3, u_1 \rangle}{\langle u_1, u_1 \rangle} \cdot u_1 - \frac{\langle v_3, u_2 \rangle}{\langle u_2, u_2 \rangle} \cdot u_2 = \begin{pmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$q_1 = \frac{u_1}{\|u_1\|} = \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} \quad q_2 = \frac{u_2}{\|u_2\|} = \begin{pmatrix} -2/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{pmatrix} \quad q_3 = \begin{pmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$A = \begin{bmatrix} \underbrace{1}_{v_1} & \underbrace{0}_{v_2} & \underbrace{0}_{v_3} \\ \underbrace{1}_{v_1} & \underbrace{1}_{v_2} & \underbrace{0}_{v_3} \\ \underbrace{1}_{v_1} & \underbrace{1}_{v_2} & \underbrace{1}_{v_3} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & -2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix} \cdot \begin{bmatrix} \sqrt{3} & 2/\sqrt{3} & 0 \\ 0 & \sqrt{6}/3 & 0 \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix}$$

$q_1 \quad q_2 \quad q_3$

$$V \rightarrow U \rightarrow Q$$

ортон. ортонорм.

$$\begin{bmatrix} \|u_1\| & \frac{\langle v_2, u_1 \rangle}{\|u_1\|} & \frac{\langle v_3, u_1 \rangle}{\|u_1\|} \\ 0 & \|u_2\| & \frac{\langle v_3, u_2 \rangle}{\|u_2\|} \\ 0 & 0 & \|u_3\| \end{bmatrix}$$

$$A = Q \cdot R$$

$$\begin{bmatrix} A_1 & \dots & A_n \end{bmatrix}_{[m \times n]} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}_{[m \times k]}^Q \cdot \begin{bmatrix} & & \\ & * & \\ 0 & & * \end{bmatrix}_{[k \times n]}^R$$

$k < n$

лин. зав.
 части

Q - матрица с ортонорм. столбцами
 R - верхнетреугольная

$$Q^T Q = I$$

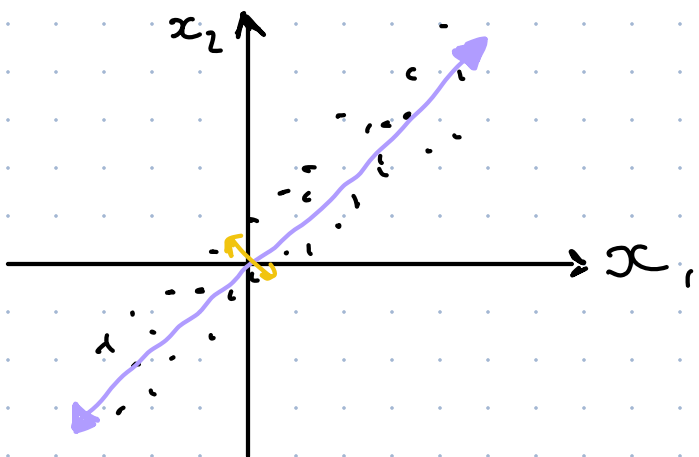
$[k \times n] [n \times k]$

Умберштейн:

QR-разложение
матрицы,

∃ для любой прямоугольной

PCA:



$$z_1 = L_{11}x_1 + L_{12}x_2$$

$$\text{Var}(z_1) \rightarrow \max_{L_{11}, L_{12}}$$

$$z_2 = L_{21}x_1 + L_{22}x_2$$

$$\text{Var}(z_2) \rightarrow \max_{L_{21}, L_{22}}$$

$$z_2 \perp z_1$$

$$X = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_k \\ 1 & 1 & \dots & 1 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 1 & \dots & 1 \\ z_1 & z_2 & \dots & z_k \\ 1 & 1 & \dots & 1 \end{bmatrix}$$

$$\text{span}\{x_1, \dots, x_k\} = \text{span}\{z_1, \dots, z_k\}$$

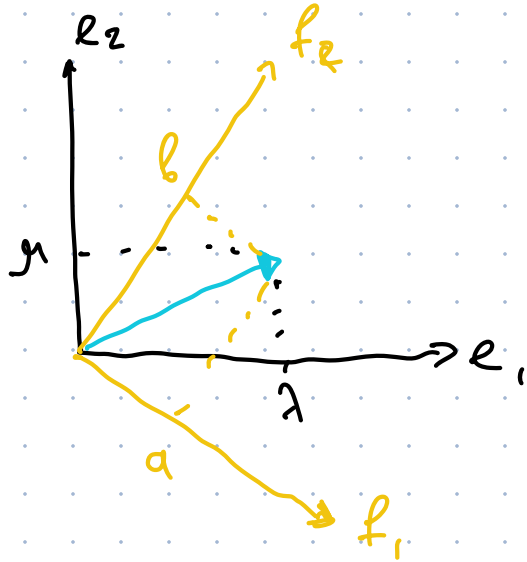
$$Z = X \cdot W$$

$$ZW^T = X$$

~~QR~~

⑤ Смена базиса

(λ, μ) — коорд. вектор v в e_1, e_2
 (a, b) f_1, f_2



$$f_1 = c_{11} e_1 + c_{21} e_2$$

$$f_2 = c_{12} e_1 + c_{22} e_2$$

$$\begin{pmatrix} f_1 & f_2 \end{pmatrix} = \begin{pmatrix} e_1 & e_2 \end{pmatrix} \cdot \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}$$

$$v = \lambda e_1 + \mu e_2 = (e_1, e_2) \begin{pmatrix} \lambda \\ \mu \end{pmatrix}$$

$$v = a f_1 + b f_2 = (f_1, f_2) \begin{pmatrix} a \\ b \end{pmatrix}$$

$$(f_1, f_2) \begin{pmatrix} a \\ b \end{pmatrix} = (e_1, e_2) \begin{pmatrix} \lambda \\ \mu \end{pmatrix}$$

~~$$\begin{pmatrix} e_1 & e_2 \end{pmatrix} \cdot \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} e_1 & e_2 \end{pmatrix} \begin{pmatrix} \lambda \\ \mu \end{pmatrix}$$~~

$$C \cdot \underset{y}{\begin{pmatrix} a \\ b \end{pmatrix}} = \underset{x}{\begin{pmatrix} \lambda \\ \mu \end{pmatrix}}$$

$$x = C y$$

$$y = C^{-1} x$$